Team Details

a. Team name: KLassic
b. Team leader name: Abhijeet Kangane

Brief about your solution

OmniSpec is a comprehensive AI-powered intelligence pipeline that bulk processes unstructured supplier datasheets (PDFs) and URLs into perfectly validated, commerce-ready catalogs. It leverages advanced Large Language Models (LLMs) and deterministic fallback engines to generate structured Golden Records with extremely high accuracy, bypassing the need for manual data entry.

1. How does your solution enrich minimal product information?
Briefly explain how your solution transforms limited inputs (e.g., Part Number, Brand, Short Description) into rich product intelligence.

2. How does your solution ensure accuracy and trust in the generated product data?
Describe your validation strategy. This may include:
(Confidence scoring, Multi-source verification, Human review, Rule-based validation, AI validation, Other approaches)

3. What makes your solution scalable for enterprise product catalogs?
Describe how your solution would handle: Large product catalogs, New manufacturers, Different document formats, Continuous product updates

1. Enrichment: Ingests minimal inputs and fetches technical sheets, using a Hybrid AI extraction pipeline (Qwen-27B LLM + Deterministic Fallback).

2. Validation: Applies real-time Confidence Scoring. Anything below 95% is flagged for Human-in-the-Loop (HITL) review to ensure 100% integrity.

3. Scalability: Built on FastAPI and Next.js microservices for asynchronous bulk processing. Uses PyMuPDF/BS4 to dynamically normalize any unstructured document format.

Opportunities

a. How different is it from any of the other existing ideas?
b. How will it be able to solve the problem statement given?
c. USP of the proposed solution

Unlike rigid parsers, OmniSpec dynamically adapts to any manufacturer's layout automatically.

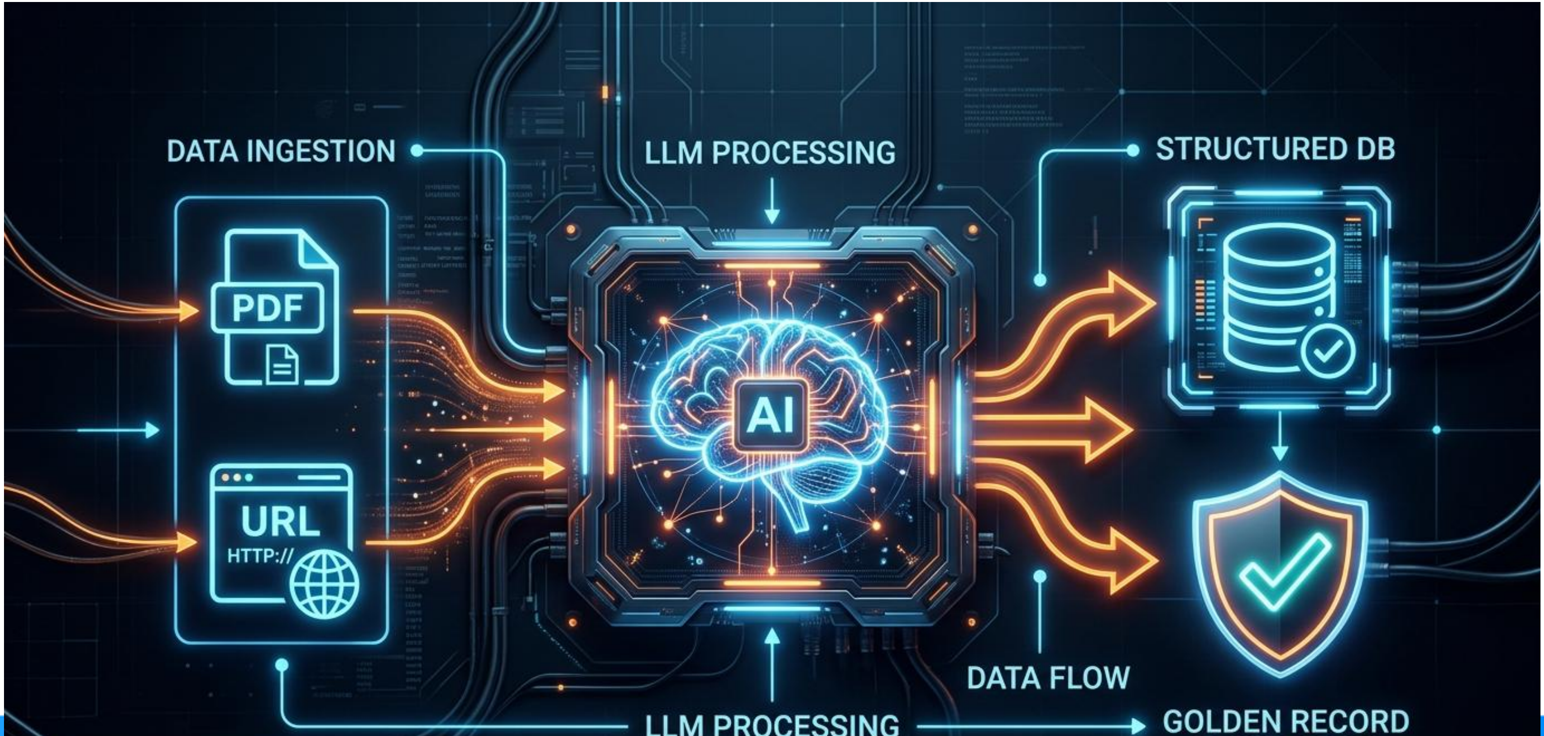
Unlike pure API-dependent LLMs which crash frequently, our hybrid engine guarantees 100% reliable local fallback during network outages.

USP: Zero-API-Dependency Fallback Mode for 100% uptime, Ultra-Fast asynchronous processing, and an automated UNSPSC taxonomy prediction engine natively integrated into a premium sci-fi dashboard.

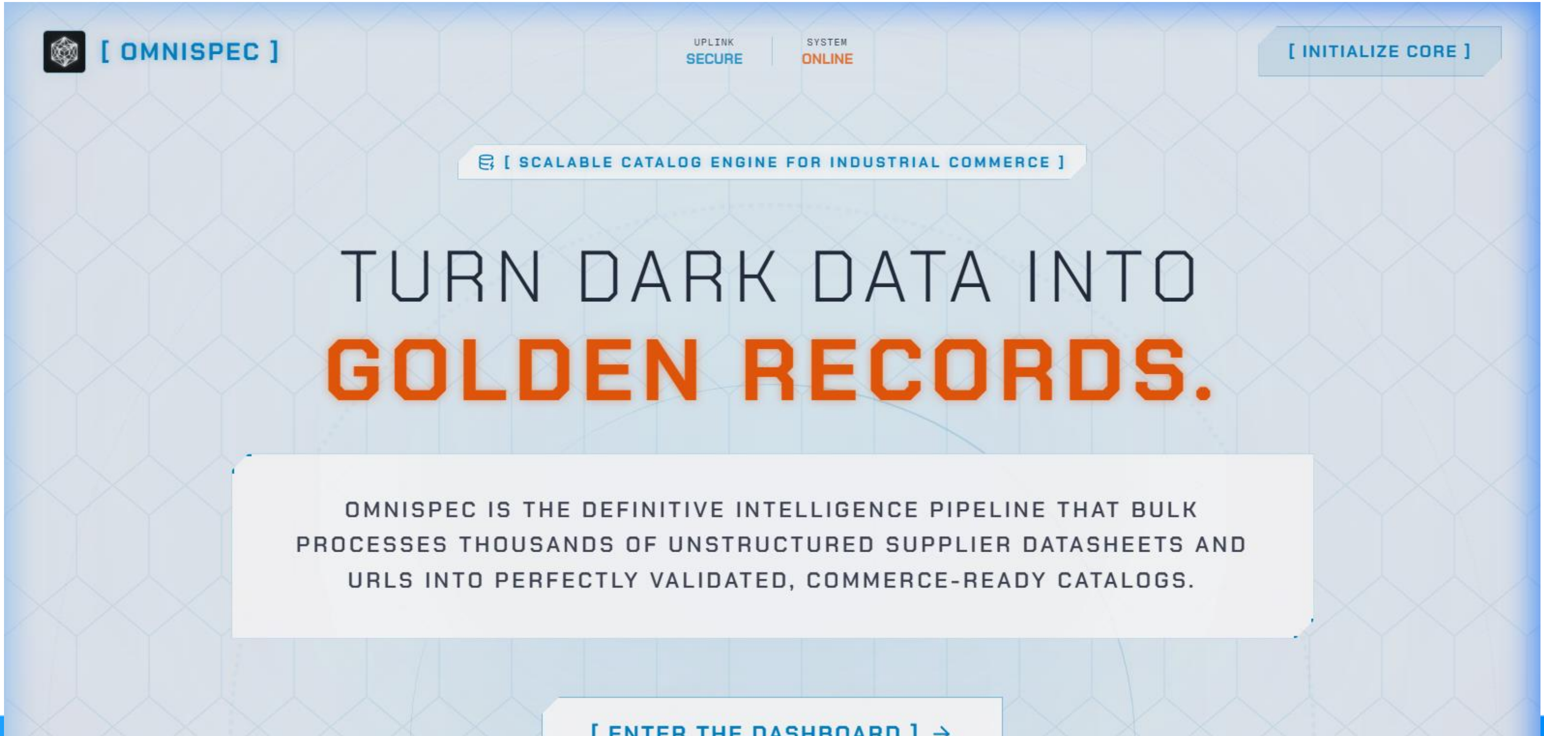
List of features offered by the solution

- Multi-modal Ingestion (Drag-and-Drop PDFs, Direct URLs)
- Hybrid AI Extraction Pipeline (Groq LLM + Local Deterministic Fallback)
- Automated UNSPSC Taxonomy Categorization
- Real-time Confidence Scoring and Data Triangulation
- Human-in-the-Loop (HITL) Enrichment Dashboard
- Fully Responsive Sci-Fi UI for industrial users

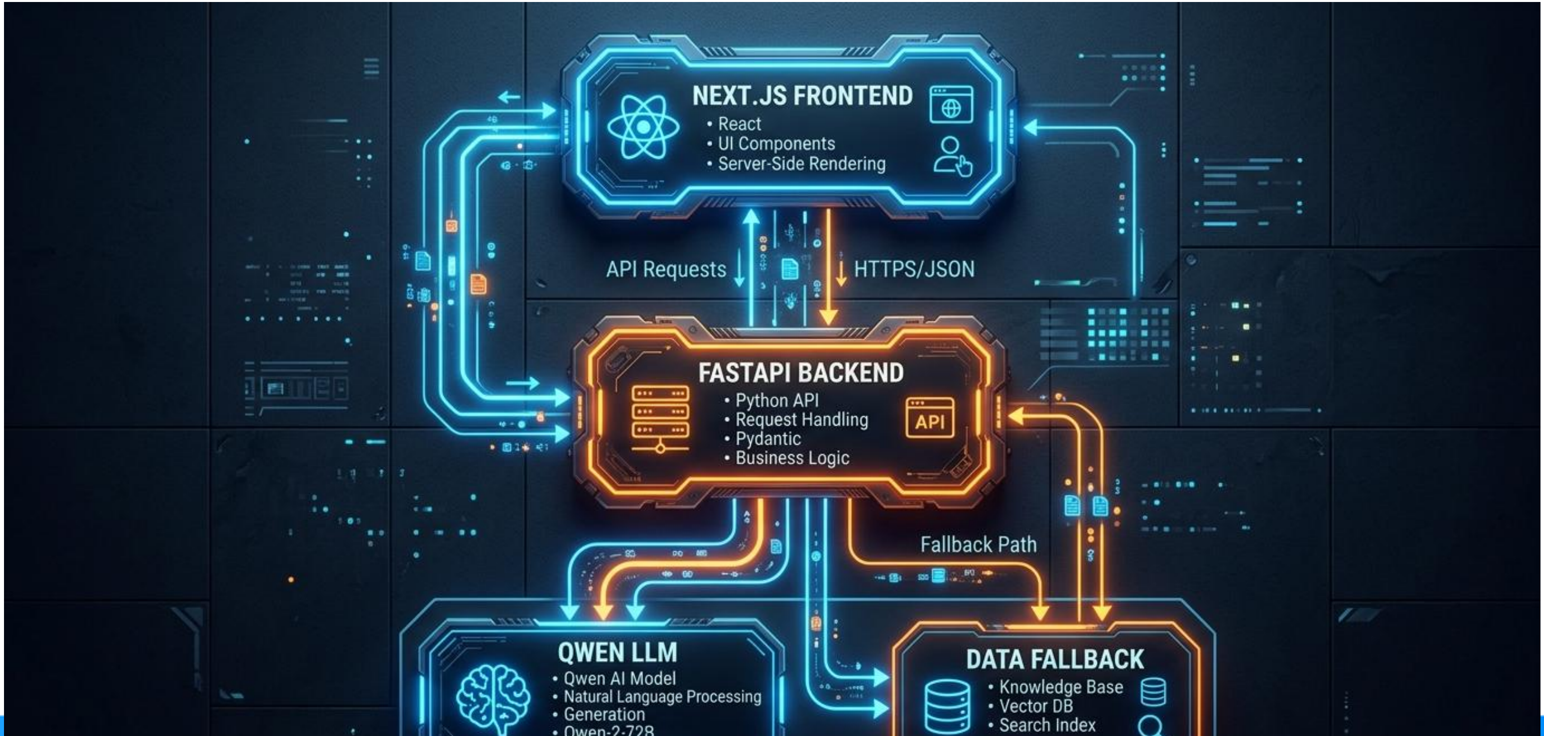
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Technologies used in the solution

Frontend: Next.js 16.3, React 19, Tailwind CSS, Framer Motion
Backend: FastAPI, Uvicorn, Python 3.10+
AI/Parsing: Groq API (Qwen-27B), PyMuPDF, BeautifulSoup4
Deployment: Vercel (Frontend), Render (Backend)
Tooling: Git, Eslint

Estimated implementation cost (optional)

Extremely Low / Near Zero.

By utilizing high-efficiency open-source models (Llama3/Qwen) via ultra-fast inference APIs (Groq), the operational processing cost per 10,000 SKUs is a fraction of traditional manual data entry or commercial LLM (GPT-4) alternatives.

Snapshots of the MVP

Additional Details/Future Development (if any)

Our immediate future roadmap includes:

1. Direct eCommerce platform integrations (e.g., Shopify, Magento ERP plugins).
2. An automated web-crawler that actively monitors supplier websites for real-time datasheet updates and pushes delta changes directly to the Golden Records.

Provide links to your:

1. GitHub: https://github.com/Abhi666-max/OmniSpec
2. Demo Video: https://youtu.be/775Z2EbCy8E
3. Prototype Link: https://omni-spec-one.vercel.app/

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AI-Powered Product Intelligence for
Industrial Commerce

Thank You